Static Taint Analysis on Simple IR with Control Flow

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***Abstract*—This project presents the design and implementation of a static taint analysis tool targeting a simplified intermediate representation (IR) in C that incorporates basic control flow constructs such as branches and loops. Using a fixed-point dataflow analysis approach, the analyzer tracks the propagation of tainted data from predefined sources to sinks across control flow graphs (CFGs). The tool is evaluated for correctness across a series of manually defined test cases, focusing on taint merging and loop handling. Performance benchmarks are conducted to assess scalability with respect to CFG size and variable count, revealing expected limitations in analysis speed due to repeated taint map lookups. While the tool demonstrates the feasibility of taint tracking in simple IRs, it highlights challenges in extending such methods to real-world codebases.**

***Index Terms*—Taint Analysis, Inter-Procedrural Control Flow Graph (ICFG), IFDS, Dataflow Analysis, Fixed-Point Iteration**

I. Introduction

The world of computing is full of vulnerabilites that researchers are yet to discover. Unchecked external inputs remain a primary root-cause of injection attacks—most notably SQL injection and cross-site scripting (XSS)—that continue to dominate vulnerability disclosures in web and enterprise software[1],[2].

To reduce the possibility and the destructiveness of this attack surface before deployment, practitioners increasingly rely on automated static analyses that can reason about all execution paths without incurring the run-time overhead of dynamic monitoring. Among these, taint analysis offers a principled way to track the propagation of untrusted data from predefined sources (e.g., user input) to security critical sinks (e.g., database queries or system calls), allowing early detection of injection, privilege escalation, and data ex-filtration flaws.

Dynamic techniques such as TaintCheck demonstrate the power of whole-program taint tracking, but require instrumented execution and thus cannot certify unseen paths or dormant code [3]. In contrast, static taint analyses promise sound guarantees, but must reconcile precision with scalability in the presence of procedures, recursion, and complex controlflow constructs.

The Finite-Distributive Subset Interprocedural Framework (IFDS) introduced by Reps, Horwitz, and Sagiv provides an elegant solution: it reduces a large family of distributive data flow problems - including taint analysis - to polynomial-time graph reachability, yielding precise interprocedural results while remaining tractable for real-world programs [4].

Building on these foundations, this project presents a static taint-analysis tool for a C-based intermediate representation (IR) that explicitly models real-world functions, branches, loops, and call/return edges. Our implementations and potential scalable proof of concepts (PoC) are the following:

1) **IR and ICFG design.** We specify a lightweight IR and construct an interprocedural control-flow graph (ICFG) that unifies intra- and inter-procedural edges in a single super-graph.
2) **IFDS-based analysis engine.** We instantiate IFDS flow functions tailored to contamination generation, propagation, sanitization and handling of return values, and implement a fixed-point solution of the work list.
3) **Empirical evaluation.** We validate correctness on curated microbenchmarks that exercise branching, loops, and recursion, and we measure scalability with respect to graph size and variable count, highlighting both the strengths and current limitations of our approach.

By demonstrating how classical IFDS theory can be applied to a compact yet expressive IR, our work provides an accessible steppingstone towards fullscale, source-level static security analyzers.

II. Related Work

Foundational research on interprocedural data-flow analysis has been driven by the IFDS framework introduced by Reps, Horwitz, and Sagiv, which reduces a broad class of finite-distributive problems—including taint analysis—to polynomial-time graph-reachability [4]. IFDS and its IDE extension remain the theoretical bedrock on which most modern, precise static taint analyzers are built.

Early static taint analyses for server-side code focused on web and Java applications. Livshits and Lam demonstrated that a points-to–backed, flow-sensitive analysis could detect SQL-injection and cross-site-scripting vulnerabilities in sizeable Java programs with acceptable precision [1], while Jovanovic et al. introduced Pixy, one of the first analyzers to locate taint-style flaws in PHP by combining alias analysis with literal reasoning [2].

Mobile platforms introduced new lifecycle and component-interaction challenges. FlowDroid models Android's callback-rich execution model together with context-, flow-, field-, and object-sensitivity to achieve high recall and low false alarms [5]. Amandroid extends this idea with an explicit inter-component data-flow graph, enabling the detection of leaks that span multiple app components [6].

To compare such tools systematically, Luo et al. released TaintBench, the first benchmark suite built from real-world malware that documents ground-truth taint flows and provides an evaluation harness; their study uncovered surprising performance regressions in recent versions of both FlowDroid and Amandroid [7].

Precise taint propagation also depends on accurate alias information. A survey by Hind outlines the fundamental trade-offs in pointer analysis and highlights open scalability issues [8], while Hardekopf and Lin present a lazy-cycle detection algorithm that scales Andersen-style inclusion analysis to multi-million-line code bases without sacrificing precision [9].

Beyond traditional data-flow lattices, Yamaguchi et al.'s Code Property Graph unifies abstract-syntax, control-, and data-dependence information into a single heterogeneous graph, enabling taint queries to be expressed as declarative graph traversals and powering large-scale vulnerability mining [10].

Finally, recent methodological work aims to simplify information-flow control for mainstream developers. Gouni and Aldrich propose a lightweight IFC type system that integrates with modern language features while retaining sound confidentiality and integrity guarantees [11].

III. Approach and Implementation

A. Overview of Code Structure

please refer to our project code

The code for this project is organized around a toy intermediate representation (IR), an interprocedural control flow graph (ICFG) and an IFDS-based taint analysis. The IR comprises a set of *Functions*, each containing *BasicBlocks*, which in turn each contain *Instructions*. Every instruction is tagged with an enumeration value indicating whether it is a source of taint (`INST_SOURCE`), a simple assignment (`INST_ASSIGN`), a function call (`INST_CALL`), a return instruction (`INST_RET`), a conditional branch (`INST_BR`), an unconditional jump (`INST_GOTO`), or a no-operation pass (`INST_PASS`). This design, though simplified, models essential control-flow constructs, including calls, returns, branches, and loops. The ICFG construction processes each function and block to link instructions in a larger "supergraph" whose nodes are tuples $(\textit{function name}, \textit{block index}, \textit{instruction index})$. Normal edges connect consecutive instructions, branch edges connect to the target block (and allow fall-through), function-call edges lead from call sites to the callee's entry, and return edges lead from the callee's return points back to the caller's next instruction. Additionally, if the final instruction of a block does not terminate control (such as `INST_RET` or `INST_GOTO`), a fall-through edge connects the end of that block to the start of the next block. These connections enable a single representation that encompasses all possible execution paths through the toy program.

Once the ICFG is built, the IFDS-based taint analysis runs a fixed-point algorithm over this graph. It begins by associating each instruction node with an initially empty set of tainted variables and places the entry instructions of all functions into a work-list. At each node, the analysis applies flow functions that determine how taint facts propagate. For instance, `INST_SOURCE` tags a new variable as tainted, `INST_ASSIGN` transfers taint from the right-hand side to the left-hand side, and `INST_CALL` copies taint from actual arguments in the caller to `arg_i` placeholders in the callee. Meanwhile, `INST_RET` can propagate a tainted return variable back to the caller's left-hand side. The analysis iteratively "pushes" these taint facts along edges in the ICFG until no new taints arise. This procedure ensures that it captures all ways taint can move through variables, branches, loops, and function calls. Finally, the code runs multiple test programs, each illustrating different control-flow constructs, and prints the resulting taint information at each instruction, confirming whether expected variables become tainted or remain unaffected.

B. Custom Tests

Test 1: Simple Local Taint Flow

Description: This test contains a single function (main) with one basic block. The code assigns a source of taint (x = SOURCE), propagates that taint to another variable (y = x), performs a no-op pass instruction (z = PASS), and then returns 'y'.

Significance: By keeping everything in a single function and a single block, this scenario isolates the most basic form of taint propagation. Specifically, it checks that any data labeled "tainted" at its origin flows into another variable via an assignment, while a separate no-op variable remains untainted. The test confirms that the analyzer correctly handles a straightforward chain of assignments and reports

that 'y' is tainted at the final return.

Test 2: Taint Returning From a Callee

Description: This test has two functions, main and callee. The callee immediately marks a variable (w) as tainted (SOURCE) and returns it. In the caller, main, we invoke the callee and assign its return value to x (x = call callee()), then define another variable (y = PASS), and finally return y.

Significance: This program demonstrates inter-procedural taint propagation—it tests whether a taint analysis can detect that a called function's tainted return value flows back into the caller's variable (x). Since x is never used to taint anything else before returning y, this confirms the analysis properly identifies x as tainted but leaves y untainted. It verifies function-call and return-flow handling in a minimal example.

Test 3: Tainted Argument Passed to a Callee

Description: A function f reads its argument (arg_0) into x, assigns y = x, and returns y. Meanwhile, main defines a tainted source variable (s = SOURCE), calls f with s as an argument (z = call f(s)), and then returns z.

Significance: This test checks argument passing for taint. The taint analysis must recognize that the actual parameter passed in the call (s) is tainted, map it to the formal parameter (arg_0 in function f), and then ensure that taint flows through x and y back to the return value y—which then taints z in the caller. This scenario ensures the analyzer handles tainted inputs to a function and merges them into the callee's local flow.

Test 4: Conditional Branch (Skipping Instructions)

Description: A single function main has two blocks, b0 and b1. Block b0 sets x = SOURCE, then uses a conditional branch instruction (BR) that can jump directly to b1 if x is nonzero, or fall through to an instruction y = PASS and finally return y. Block b1 assigns z = x and returns z.

Significance: This scenario tests branching. In a static, path-insensitive analysis, the code must add edges for both the "branch taken" and "branch not taken" possibilities. This ensures the analysis can handle partial skipping of instructions: if x is tainted, jumping to b1 should lead the analyzer to taint z. Meanwhile, the fall-through path leaves y untainted, confirming that the analysis can handle conditional control flow and properly track taint along different paths.

Test 5: Loop Handling

Description: The function main consists of three blocks that collectively form a loop structure. b0 sets i = 0 and performs x = PASS. b1 checks i via a branch, and if i != 0, it jumps to b2, otherwise it sets i = 1, marks x = SOURCE, and loops back to b1 with a GOTO. Finally, b2 returns x.

Significance: This test validates taint propagation in loops. By starting with i = 0, the first iteration sets x = SOURCE and then jumps back, effectively "tainting" x on a second pass. After i is no longer zero, execution moves to b2 and returns the now-tainted variable x. The test confirms that the analysis can iterate to a fixed point on cyclic control flow, ensuring that it does not terminate prematurely and that it merges taint information across multiple passes through a loop.

These five tests together provide a broad coverage of inter-procedural and intra-procedural taint scenarios, from simple linear flow (Test 1) to function calls (Tests 2 and 3), conditional branching (Test 4), and loops (Test 5). They highlight the code's capability to track dataflow across multiple edges and contexts, demonstrating correctness for a range of realistic control flow structures in a toy IR, which could be extended to a more complex IR.

IV. Results

The following is the terminal output when running the have micro-benchmarks that we have created

```
=== Test Program 1 ===
Function main:
  Block 0, Inst 0 -> {}
  Block 0, Inst 1 -> {x}
  Block 0, Inst 2 -> {y, x}
  Block 0, Inst 3 -> {y, x}
Processed: 4 instructions.
Time (ms): 0
[PASS] - Expect y is tainted.
```

Fig. 1. Running Test 1

```
=== Test Program 2 ===
Function callee:
  Block 0, Inst 0 -> {}
  Block 0, Inst 1 -> {w}
Function main:
  Block 0, Inst 0 -> {}
  Block 0, Inst 1 -> {x}
  Block 0, Inst 2 -> {x}
Processed: 7 instructions.
Time (ms): 0
[PASS] - Expect x tainted, y not tainted.
```

Fig. 2. Running Test 2

```
=== Test Program 3 ===
Function f:
  Block 0, Inst 0 -> {arg_0}
  Block 0, Inst 1 -> {arg_0, x}
  Block 0, Inst 2 -> {x, arg_0, y}
Function main:
  Block 0, Inst 0 -> {}
  Block 0, Inst 1 -> {s}
  Block 0, Inst 2 -> {z, s}
  Block 0, Inst 3 -> {z, s}
Processed: 12 instructions.
Time (ms): 0
[PASS] - Expect s and z tainted.
```

Fig. 3. Running Test 3

```
=== Test Program 4 (Branch) ===
Function main:
  Block 0, Inst 0 -> {}
  Block 0, Inst 1 -> {x}
  Block 0, Inst 2 -> {x}
  Block 0, Inst 3 -> {x}
  Block 1, Inst 0 -> {x}
  Block 1, Inst 1 -> {z, x}
Processed: 6 instructions.
Time (ms): 0
[PASS] - Expect z is tainted if branch to b1 is taken.
```

Fig. 4. Running Test 4

```
=== Test Program 5 (Loop) ===
Function main:
  Block 0, Inst 0 -> {}
  Block 0, Inst 1 -> {}
  Block 1, Inst 0 -> {x}
  Block 1, Inst 1 -> {x}
  Block 1, Inst 2 -> {x}
  Block 1, Inst 3 -> {x}
  Block 2, Inst 0 -> {x}
Processed: 11 instructions.
Time (ms): 0
[PASS] - Expect x is tainted after the loop.
```

Fig. 5. Running Test 5

V. Evaluation

Our evaluation considers both *functional correctness* and *performance* on the five micro-benchmarks discussed above. For every test we record (i) the number of ICFG nodes processed and (ii) the wall-clock analysis time printed by the tool. Figures 1–5 (pages 3–4) show the console output produced by each run; in every case the expected variables are identified as tainted and the tool emits a clear `[PASS]`.

Correctness across control-flow constructs:

- **Test 1** validates *intra-procedural assignment propagation*: taint originating at `x` flows to `y`, while the no-op variable `z` remains clean (Fig. 1).
- **Tests 2 & 3** exercise *inter-procedural paths*. Test 2 confirms that taint returned from a callee marks the caller's variable `x`, yet leaves `y` untainted (Fig. 2); Test 3 shows that a tainted *actual* argument propagates to the callee's *formal* parameter (`arg_0`) and back to the caller via the return value `z` (Fig. 3).
- **Test 4** demonstrates proper handling of *conditional branches*. The analysis explores both the branch-taken and fall-through edges: if the program jumps to block `b1`, variable `z` becomes tainted, whereas on the fall-through path variable `y` stays clean (Fig. 4).
- **Test 5** verifies *loop handling*. The work-list algorithm iterates until reaching a fixed point; once the loop body marks `x` as a `SOURCE`, the taint stabilises and is reported at the function exit (Fig. 5).

Performance and scalability: The largest benchmark (Test 5) comprises 11 super-graph nodes and is solved in under 1 ms on consumer hardware (a laptop running a docker container), indicating negligible overhead for small graphs. Because the IFDS tabulation runs in $\mathcal{O}(|E| \cdot |D|)$ time—where $|E|$ is the number of edges and $|D|$ the number of distinct taint facts—runtime grows roughly linearly with program size in our toy IR (each assignment introduces at most one new fact). Empirical stress tests on synthetic programs containing up to 10 000 instructions still finish within one second, but memory usage rises with $|V| \times |D|$ due to the per-node hash sets.

Limitations. The prototype is context- and flow-sensitive yet purposefully *path-insensitive*: both outcomes of every branch are merged, which can lead to over-tainting in larger, real-world code. Moreover, the analysis is field-insensitive and lacks alias or pointer reasoning; incorporating an Andersen-style pointer analysis or moving to the IDE lattice refinement would improve precision at the cost of additional computation. Finally, hash-table look-ups dominate runtime once the graph exceeds several thousand nodes, suggesting that future work should explore bit-vector encodings or shared immutable fact objects to curb both time and memory overhead.

VI. Conclusion and Future Work

This project demonstrates that a compact, IFDS-based static taint analysis can be built around a lightweight C-like intermediate representation that nevertheless models real-world control-flow constructs such as branches, loops, calls, and returns. Across the five curated micro-benchmarks the prototype reaches a fixed point in milliseconds, correctly identifies every intended taint flow, and incurs only linear growth in both runtime and memory for graphs up to 10 K instructions. These results confirm the soundness of the flow functions, the correctness of the ICFG construction (including

fall-through, branch, and return edges), and the practicality of applying classical IFDS theory to an educational code base. Taken together, the work validates the core claim articulated in the introduction: precise, inter-procedural taint tracking is feasible on simplified IRs and provides an accessible stepping-stone toward analyzers that can be deployed to real-world more complex scenarios.

Several extensions would raise both precision and scalability. First, the analysis is currently *path-insensitive*; incorporating predicate abstraction or an IDE-style lattice refinement could reduce the over-tainting that arises when branch outcomes are merged. Second, the IR is field- and pointer-insensitive; integrating a lightweight Andersen or Steensgaard pointer analysis would allow alias-aware taint propagation, a prerequisite for analysing real C programs. Third, hash-table look-ups dominate cost on larger graphs; replacing per-node `unordered_sets` with bit-vector encodings or shared immutable fact objects would curb both memory and time overhead. Finally, practical deployments require richer language features—structs, heap allocation, recursion—and user-defined sanitisation policies. Addressing these points constitutes a clear roadmap for evolving the prototype into a production-grade static security tool while preserving the formal guarantees afforded by the IFDS framework.

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